Recognizing clothing items

April 11, 2026

[1]:
```
from tensorflow import keras
```

[2]:
```
(train_images, train_labels), (test_images, test_labels) = keras.datasets.
↪fashion_mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-
datasets/train-labels-idx1-ubyte.gz
29515/29515              0s 1us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-
datasets/train-images-idx3-ubyte.gz
26421880/26421880              1s
0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-
datasets/t10k-labels-idx1-ubyte.gz
5148/5148              0s 1us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-
datasets/t10k-images-idx3-ubyte.gz
4422102/4422102              0s
0us/step
```

[3]:
```
import tensorflow as tf
```

[4]:
```
import numpy as np
```

[5]:
```
from matplotlib import pyplot as plt
```

[6]:
```
class_names = ['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat',
               'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle boot']
```

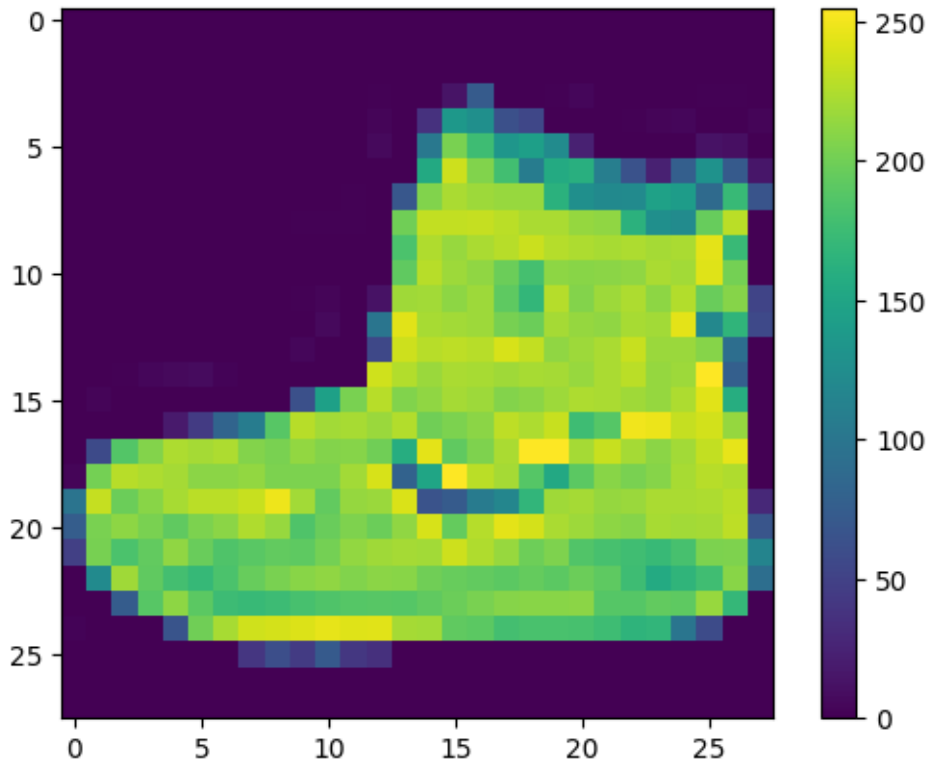
[7]:
```
train_images.shape
```

[7]: (60000, 28, 28)

[8]:
```
test_images.shape
```

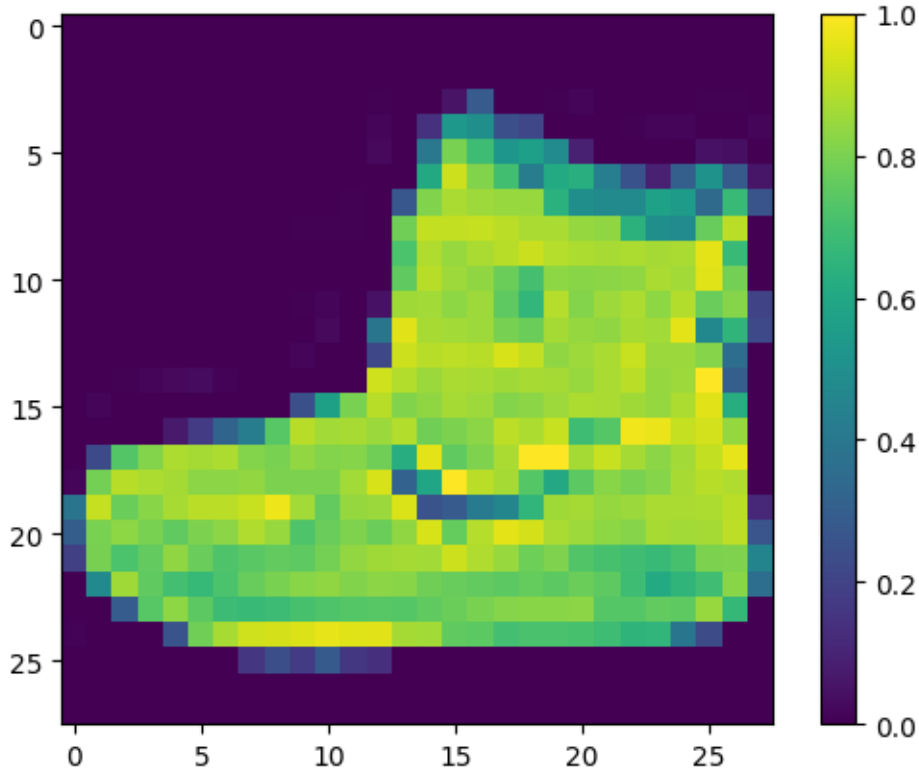
[8]: (10000, 28, 28)

[9]:
```
plt.imshow(train_images[0])
plt.colorbar()
plt.grid(False)
```



[10]:
```
train_images = train_images / 255.0
test_images = test_images / 255.0
```

[11]:
```
plt.imshow(train_images[0])
plt.colorbar()
plt.grid(False)
```



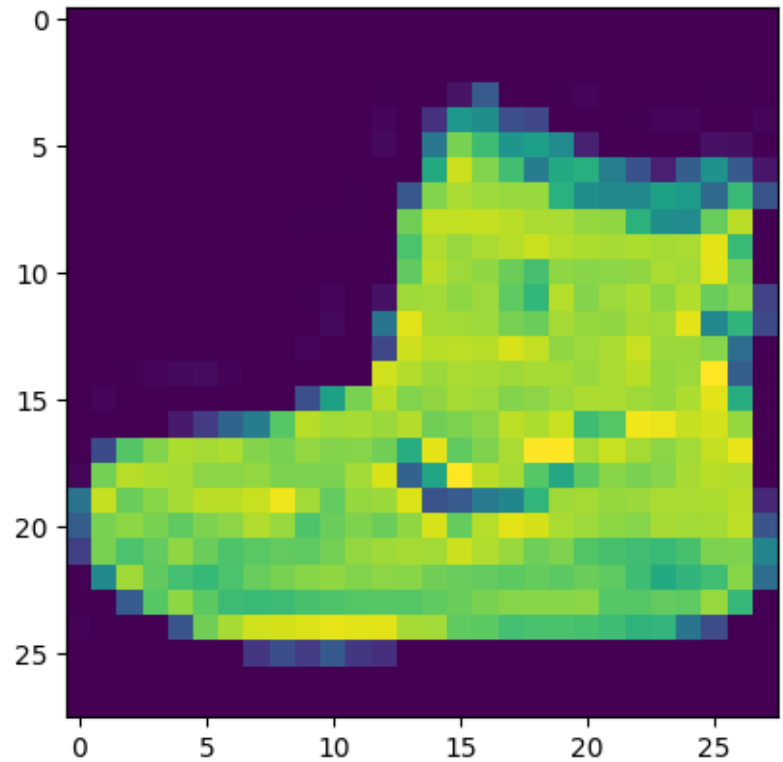
[12]:
```
plt.figure(figsize=(10, 10))
for i in range(25):
    plt.subplot(5, 5, i+1)
    plt.xticks([])
    plt.yticks([])
    plt.grid(False)
    plt.imshow(train_images[i], cmap=plt.cm.binary)
    plt.xlabel(class_names[train_labels[i]])
plt.show()
```

[21]:
```
#import skimage.transform
from skimage.transform import pyramid_gaussian


def get_pyramid_features(img):
    return np.hstack([
        layer.reshape(-1)
        #for layer in skimage.transform.pyramids.pyramid_gaussian(img)
        for layer in pyramid_gaussian(img, max_layer=3, downscale=2)
    ])
```

[22]:
```
from skimage import data, feature, exposure
```

[23]:
```
plt.imshow(train_images[0])
```

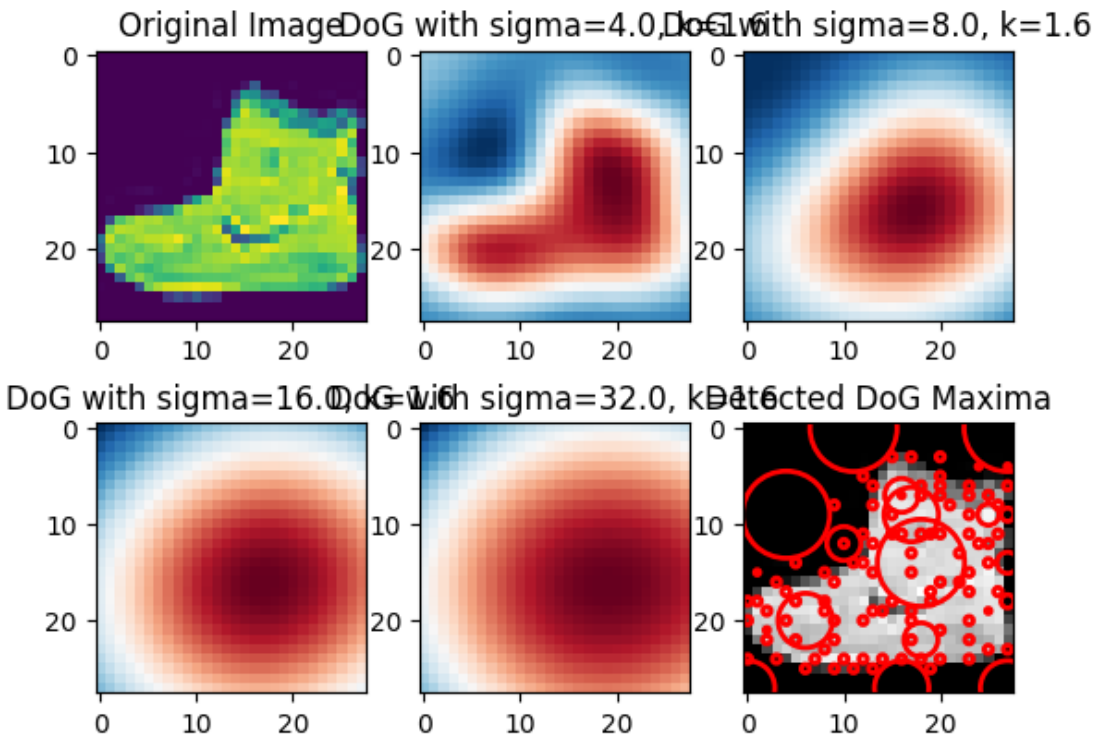
[23]: <matplotlib.image.AxesImage at 0x15175c45950>



[24]:
```
from skimage import data, feature, color, filters, img_as_float
from matplotlib import pyplot as plt


original_image = img_as_float(train_images[0])
#img = color.rgb2gray(original_image)
img = original_image

k = 1.6

plt.subplot(2,3,1)
plt.imshow(original_image)
plt.title('Original Image')


sigmas = np.array([4.0, 8.0, 16.0, 32.0])
for idx, sigma in enumerate(sigmas):
    s1 = filters.gaussian(img,k*sigma)
```

```
    s2 = filters.gaussian(img,sigma)

    # multiply by sigma to get scale invariance
    dog = s1 - s2
    plt.subplot(2,3,idx+2)
    print(dog.min(),dog.max())
    plt.imshow(dog,cmap='RdBu')
    plt.title('DoG with sigma=' + str(sigma) + ', k=' + str(k))

ax = plt.subplot(2,3,6)

def blobify(img):
    blobs_dog = [
        (x[0], x[1], x[2])
        for x in feature.blob_dog(
            img, min_sigma=1e-12, max_sigma=5, threshold=0.0001, overlap=1.0
        )
    ]
    blobs_dog += [
        (x[0], x[1],x[2])
        for x in feature.blob_dog(
            -img, min_sigma=1e-12, max_sigma=5, threshold=0.0001, overlap=1.0
        )
    ]
    return blobs_dog


img_blobs = color.gray2rgb(img)
for blob in blobify(img):
    y, x, r = blob
    c = plt.Circle((x, y), r, color='red', linewidth=2, fill=False)
    ax.add_patch(c)
plt.imshow(img_blobs)
plt.title('Detected DoG Maxima')

plt.show()
```

```
-0.20447270138324025 0.0993683816955817
-0.21556082223199366 0.042726124349179974
-0.09428870469466712 -0.014654777734303834
-0.03023575252106837 -0.019981221057295676
```



[25]:
```
import numpy as np
from sklearn.svm import LinearSVC
from tqdm.notebook import tqdm

import time

print(time.asctime( time.localtime(time.time()) ))
start_time = time.time()

def featurize(x_train, y_train):
    data = []
    labels = []
    for img, label in tqdm(zip(x_train, y_train)):
        data.append(get_pyramid_features(img))
        labels.append(label)

    data = np.array(data)
    labels = np.array(labels)
    return data, labels

x_tra, y_tra = featurize(train_images, train_labels)
clf = LinearSVC(C=1, loss='hinge').fit(x_tra, y_tra)
```

```
x_val, y_val = featurize(test_images, test_labels)
print('accuracy: {:.3f}'.format(clf.score(x_val, y_val)))


end_time = time.time()
print("Execution Time: ", int(end_time - start_time),'s')
```

```
Thu Apr  9 23:16:14 2026

0it [00:00, ?it/s]

C:\Users\yathi\anaconda3\Lib\site-packages\sklearn\svm\_base.py:1258:
ConvergenceWarning: Liblinear failed to converge, increase the number of
iterations.
  warnings.warn(

0it [00:00, ?it/s]

accuracy: 0.840
Execution Time:  223 s
```

[26]:
```
import numpy as np
from tensorflow.keras.layers import Dense, Dropout, Flatten, Activation
from tensorflow.keras.layers import Conv2D, MaxPooling2D
import tensorflow as tf

def compile_model(model):
    model.summary()
    model.compile(
        optimizer='adam',
        loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
        metrics=['accuracy']
    )

def create_mlp():
    model = tf.keras.Sequential([
        tf.keras.layers.Flatten(input_shape=(28, 28)),
        tf.keras.layers.Dense(128, activation='relu'),
        tf.keras.layers.Dense(10)
    ])
    compile_model(model)
    return model


def create_lenet():
    model = tf.keras.Sequential([
        tf.keras.layers.Conv2D(filters=6, kernel_size=(5, 5), padding='valid',␣
 ↪input_shape=(28, 28, 1), activation='tanh'),  # C1
        tf.keras.layers.MaxPooling2D(pool_size=(2, 2)),
```

```
        tf.keras.layers.Conv2D(filters=16, kernel_size=(5, 5), padding='valid',␣
↪activation='tanh'),  # C3
        tf.keras.layers.MaxPooling2D(pool_size=(2, 2)),
        tf.keras.layers.Flatten(),
        tf.keras.layers.Dense(120, activation='tanh'),
        tf.keras.layers.Dense(84, activation='tanh'),
        tf.keras.layers.Dense(10, activation='softmax')
    ])
    compile_model(model)
    return model
```

[27]:
```
model = create_mlp()
```

```
C:\Users\yathi\anaconda3\Lib\site-
packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential models,
prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(**kwargs)
```

Model: "sequential"

```
 Layer (type)                    Output Shape                  ␣
↪Param #

 flatten (Flatten)               (None, 784)                     ␣
↪  0

 dense (Dense)                   (None, 128)                   ␣
↪100,480

 dense_1 (Dense)                 (None, 10)                      ␣
↪1,290
```

Total params: 101,770 (397.54 KB)

Trainable params: 101,770 (397.54 KB)

Non-trainable params: 0 (0.00 B)

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used.
Please use WSL2 or the TensorFlow-DirectML plugin.
```

[29]:
```
import numpy as np
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import ConfusionMatrixDisplay
import matplotlib
import seaborn as sns
import pandas as pd

font = {'family' : 'normal',
        'weight' : 'bold',
        'size'   : 12}
matplotlib.rc('figure', figsize=(10, 10))
np.set_printoptions(precision=2)
matplotlib.rc('font', **font)

def show_result(model, test_images):
    predictions = model.predict(test_images).argmax(axis=1)
    print(classification_report(test_labels, predictions,␣
 ↪target_names=class_names))
    cm = confusion_matrix(test_labels, predictions, normalize='pred')
    sns.heatmap(
        pd.DataFrame(cm, columns=class_names, index=class_names),
        annot=True,
        cmap='Blues',
        fmt='.2f'
    )
```

[32]:
```
def train_model(model, train_images, test_images):
    model.fit(
        train_images,
        train_labels,
        epochs=10,
        verbose=1,
        validation_data=(test_images, test_labels)
    )
    loss, accuracy = model.evaluate(test_images, test_labels, verbose=0)
    print('loss:', loss)
    print('accuracy:', accuracy)
    show_result(model, test_images)
```

[33]:
```
train_model(create_mlp(), train_images, test_images)
```

```
Model: "sequential_2"


 Layer (type)                    Output Shape                  ␣
 ↪Param #
```

```
 flatten_2 (Flatten)                     (None, 784)                                 ␣
↪  0

 dense_4 (Dense)                         (None, 128)                            ␣
↪100,480

 dense_5 (Dense)                         (None, 10)                              ␣
↪1,290
```

Total params: 101,770 (397.54 KB)

Trainable params: 101,770 (397.54 KB)

Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
1875/1875              6s 3ms/step -
accuracy: 0.8249 - loss: 0.4968 - val_accuracy: 0.8472 - val_loss: 0.4266
Epoch 2/10
1875/1875              5s 3ms/step -
accuracy: 0.8656 - loss: 0.3755 - val_accuracy: 0.8597 - val_loss: 0.3960
Epoch 3/10
1875/1875              5s 3ms/step -
accuracy: 0.8768 - loss: 0.3378 - val_accuracy: 0.8645 - val_loss: 0.3696
Epoch 4/10
1875/1875              5s 3ms/step -
accuracy: 0.8852 - loss: 0.3159 - val_accuracy: 0.8685 - val_loss: 0.3566
Epoch 5/10
1875/1875              5s 3ms/step -
accuracy: 0.8895 - loss: 0.2971 - val_accuracy: 0.8677 - val_loss: 0.3713
Epoch 6/10
1875/1875              5s 3ms/step -
accuracy: 0.8960 - loss: 0.2807 - val_accuracy: 0.8753 - val_loss: 0.3459
Epoch 7/10
1875/1875              5s 3ms/step -
accuracy: 0.9001 - loss: 0.2708 - val_accuracy: 0.8776 - val_loss: 0.3403
Epoch 8/10
1875/1875              5s 2ms/step -
accuracy: 0.9033 - loss: 0.2600 - val_accuracy: 0.8790 - val_loss: 0.3487
Epoch 9/10
1875/1875              5s 3ms/step -
accuracy: 0.9063 - loss: 0.2502 - val_accuracy: 0.8767 - val_loss: 0.3518
Epoch 10/10
```

```
1875/1875              5s 3ms/step -
accuracy: 0.9107 - loss: 0.2403 - val_accuracy: 0.8772 - val_loss: 0.3464
loss: 0.3463968336582184
accuracy: 0.8772000074386597
313/313              0s 1ms/step
              precision    recall  f1-score   support

 T-shirt/top       0.84      0.83      0.83      1000
     Trouser       0.97      0.98      0.98      1000
    Pullover       0.84      0.67      0.75      1000
       Dress       0.89      0.88      0.88      1000
        Coat       0.69      0.91      0.79      1000
      Sandal       0.98      0.96      0.97      1000
       Shirt       0.72      0.64      0.68      1000
     Sneaker       0.95      0.95      0.95      1000
         Bag       0.97      0.98      0.97      1000
  Ankle boot       0.95      0.97      0.96      1000

    accuracy                           0.88     10000
   macro avg       0.88      0.88      0.88     10000
weighted avg       0.88      0.88      0.88     10000


findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

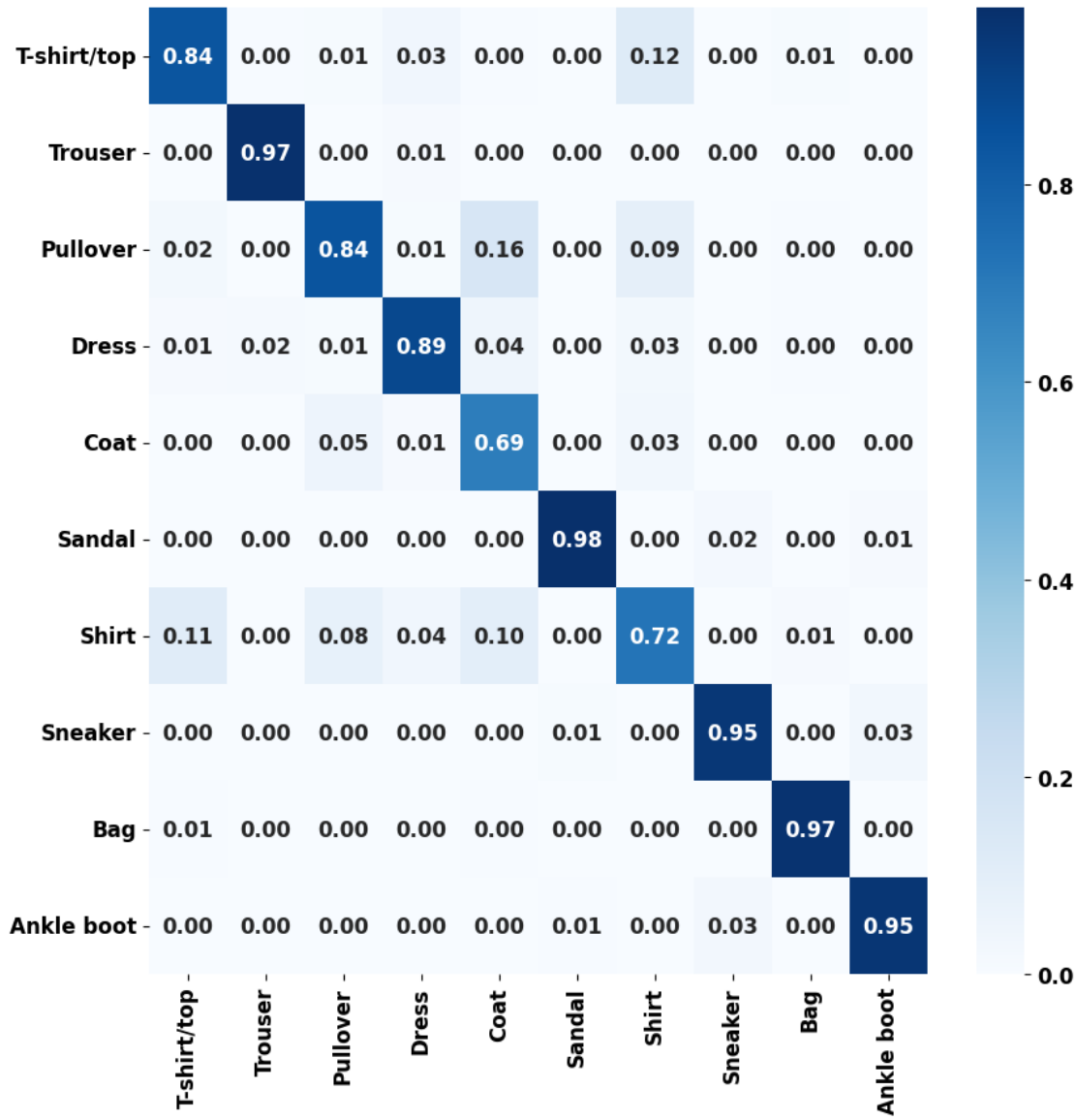
```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```



[34]:
```
train_model(
    create_lenet(),
    train_images.reshape(train_images.shape + (1,)),
    test_images.reshape(test_images.shape + (1,)),
)
```

```
C:\Users\yathi\anaconda3\Lib\site-
packages\keras\src\layers\convolutional\base_conv.py:113: UserWarning: Do not
pass an `input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in the model
instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Model: "sequential_3"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 24, 24, 6)	156
max_pooling2d (MaxPooling2D)	(None, 12, 12, 6)	0
conv2d_1 (Conv2D)	(None, 8, 8, 16)	2,416
max_pooling2d_1 (MaxPooling2D)	(None, 4, 4, 16)	0
flatten_3 (Flatten)	(None, 256)	0
dense_6 (Dense)	(None, 120)	30,840
dense_7 (Dense)	(None, 84)	10,164
dense_8 (Dense)	(None, 10)	850

```
Total params: 44,426 (173.54 KB)

Trainable params: 44,426 (173.54 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/10

C:\Users\yathi\anaconda3\Lib\site-
packages\keras\src\backend\tensorflow\nn.py:1214: UserWarning:
"`sparse_categorical_crossentropy` received `from_logits=True`, but the `output`
argument was produced by a Softmax activation and thus does not represent
logits. Was this intended?
  output, from_logits = _get_logits(
```

```
1875/1875              12s 5ms/step -
accuracy: 0.8226 - loss: 0.4875 - val_accuracy: 0.8551 - val_loss: 0.4079
Epoch 2/10
1875/1875              10s 5ms/step -
accuracy: 0.8716 - loss: 0.3489 - val_accuracy: 0.8701 - val_loss: 0.3590
Epoch 3/10
1875/1875              10s 5ms/step -
accuracy: 0.8862 - loss: 0.3091 - val_accuracy: 0.8792 - val_loss: 0.3271
Epoch 4/10
1875/1875              10s 5ms/step -
accuracy: 0.8956 - loss: 0.2830 - val_accuracy: 0.8864 - val_loss: 0.3162
Epoch 5/10
1875/1875              10s 5ms/step -
accuracy: 0.9027 - loss: 0.2625 - val_accuracy: 0.8868 - val_loss: 0.3142
Epoch 6/10
1875/1875              10s 5ms/step -
accuracy: 0.9089 - loss: 0.2456 - val_accuracy: 0.8837 - val_loss: 0.3280
Epoch 7/10
1875/1875              10s 5ms/step -
accuracy: 0.9156 - loss: 0.2299 - val_accuracy: 0.8934 - val_loss: 0.3048
Epoch 8/10
1875/1875              10s 6ms/step -
accuracy: 0.9201 - loss: 0.2172 - val_accuracy: 0.8915 - val_loss: 0.3090
Epoch 9/10
1875/1875              10s 5ms/step -
accuracy: 0.9237 - loss: 0.2054 - val_accuracy: 0.8902 - val_loss: 0.3088
Epoch 10/10
1875/1875              9s 5ms/step -
accuracy: 0.9278 - loss: 0.1936 - val_accuracy: 0.8965 - val_loss: 0.2983
loss: 0.2983458936214447
accuracy: 0.8964999914169312
313/313              1s 3ms/step

findfont: Font family 'normal' not found.

              precision    recall  f1-score   support

 T-shirt/top       0.87      0.82      0.84      1000
     Trouser       0.99      0.98      0.98      1000
    Pullover       0.83      0.84      0.84      1000
       Dress       0.90      0.89      0.89      1000
        Coat       0.84      0.83      0.83      1000
      Sandal       0.98      0.97      0.97      1000
       Shirt       0.70      0.75      0.73      1000
     Sneaker       0.94      0.95      0.95      1000
         Bag       0.97      0.98      0.97      1000
  Ankle boot       0.96      0.96      0.96      1000

    accuracy                           0.90     10000
```

```
   macro avg       0.90      0.90      0.90     10000
weighted avg       0.90      0.90      0.90     10000

findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

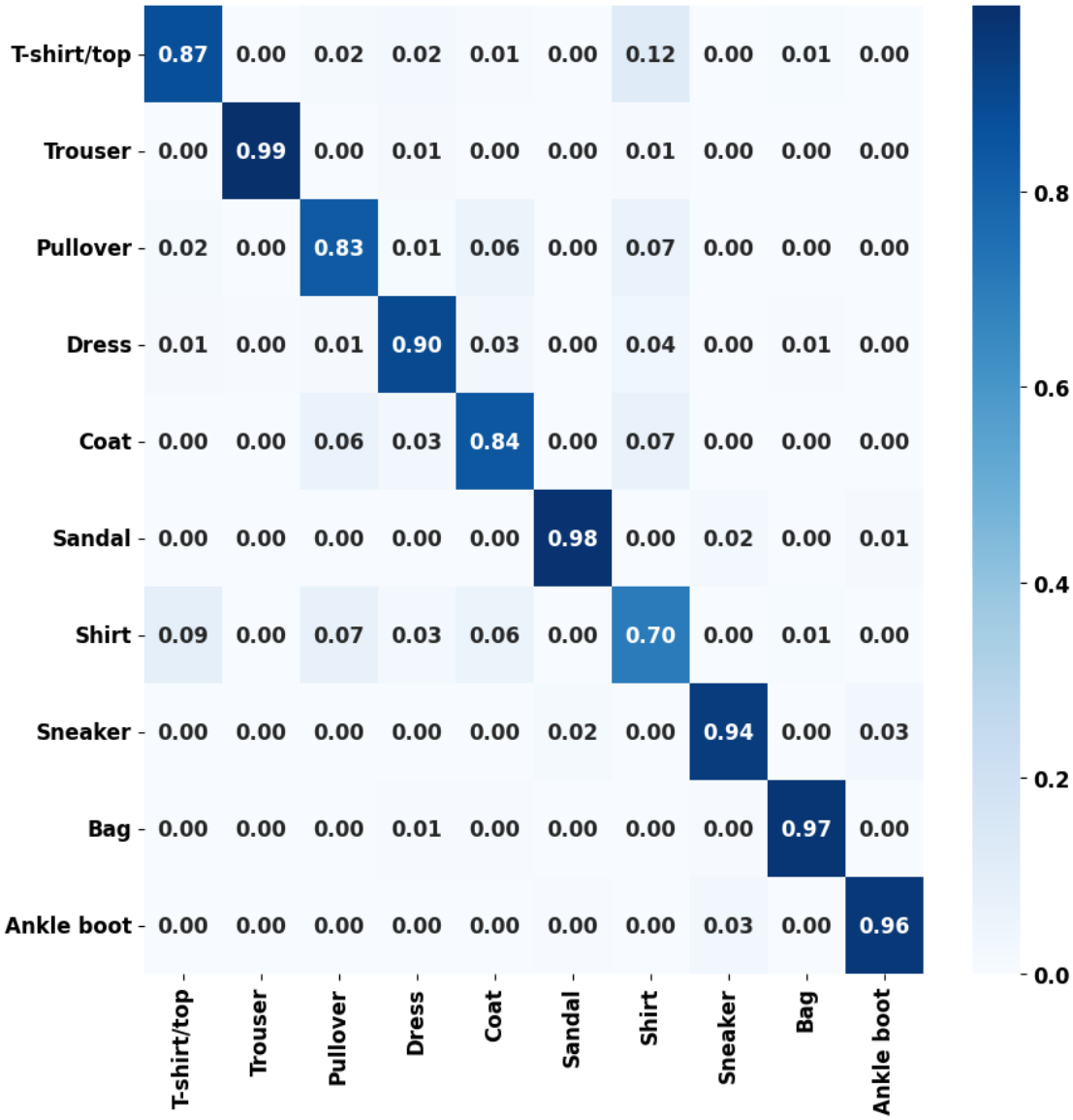
```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```

```
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
findfont: Font family 'normal' not found.
```



[13]:
```
base_model = tf.keras.applications.MobileNetV2(
    input_shape=(224, 224, 3),
    include_top=False,
    weights='imagenet'
)
```

doesn't include the classification layers

[14]:
```
base_model.summary()
```

```
Model: "mobilenetv2_1.00_224"
__________________________________________________________________________________________________
Layer (type)                    Output Shape         Param #     Connected to
==================================================================================================
input_2 (InputLayer)            [(None, 224, 224, 3) 0
__________________________________________________________________________________________________
Conv1_pad (ZeroPadding2D)       (None, 225, 225, 3)  0           input_2[0][0]
__________________________________________________________________________________________________
Conv1 (Conv2D)                  (None, 112, 112, 32) 864         Conv1_pad[0][0]
__________________________________________________________________________________________________
bn_Conv1 (BatchNormalization)   (None, 112, 112, 32) 128         Conv1[0][0]
__________________________________________________________________________________________________
Conv1_relu (ReLU)               (None, 112, 112, 32) 0           bn_Conv1[0][0]
__________________________________________________________________________________________________
expanded_conv_depthwise (Depthw (None, 112, 112, 32) 288
Conv1_relu[0][0]
__________________________________________________________________________________________________
expanded_conv_depthwise_BN (Bat (None, 112, 112, 32) 128
expanded_conv_depthwise[0][0]
__________________________________________________________________________________________________
expanded_conv_depthwise_relu (R (None, 112, 112, 32) 0
expanded_conv_depthwise_BN[0][0]
__________________________________________________________________________________________________
expanded_conv_project (Conv2D)  (None, 112, 112, 16) 512
expanded_conv_depthwise_relu[0][0
__________________________________________________________________________________________________
expanded_conv_project_BN (Batch (None, 112, 112, 16) 64
```

```
expanded_conv_project[0][0]
__________________________________________________________________________________________________
block_1_expand (Conv2D)         (None, 112, 112, 96) 1536
expanded_conv_project_BN[0][0]
__________________________________________________________________________________________________
block_1_expand_BN (BatchNormali (None, 112, 112, 96) 384
block_1_expand[0][0]
__________________________________________________________________________________________________
block_1_expand_relu (ReLU)      (None, 112, 112, 96) 0
block_1_expand_BN[0][0]
__________________________________________________________________________________________________
block_1_pad (ZeroPadding2D)     (None, 113, 113, 96) 0
block_1_expand_relu[0][0]
__________________________________________________________________________________________________
block_1_depthwise (DepthwiseCon (None, 56, 56, 96)   864
block_1_pad[0][0]
__________________________________________________________________________________________________
block_1_depthwise_BN (BatchNorm (None, 56, 56, 96)   384
block_1_depthwise[0][0]
__________________________________________________________________________________________________
block_1_depthwise_relu (ReLU)   (None, 56, 56, 96)   0
block_1_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_1_project (Conv2D)        (None, 56, 56, 24)   2304
block_1_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_1_project_BN (BatchNormal (None, 56, 56, 24)   96
block_1_project[0][0]
__________________________________________________________________________________________________
block_2_expand (Conv2D)         (None, 56, 56, 144)  3456
block_1_project_BN[0][0]
__________________________________________________________________________________________________
block_2_expand_BN (BatchNormali (None, 56, 56, 144)  576
block_2_expand[0][0]
__________________________________________________________________________________________________
block_2_expand_relu (ReLU)      (None, 56, 56, 144)  0
```

```
block_2_expand_BN[0][0]
__________________________________________________________________________________________________
block_2_depthwise (DepthwiseCon (None, 56, 56, 144)  1296
block_2_expand_relu[0][0]
__________________________________________________________________________________________________
block_2_depthwise_BN (BatchNorm (None, 56, 56, 144)  576
block_2_depthwise[0][0]
__________________________________________________________________________________________________
block_2_depthwise_relu (ReLU)   (None, 56, 56, 144)  0
block_2_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_2_project (Conv2D)        (None, 56, 56, 24)   3456
block_2_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_2_project_BN (BatchNormal (None, 56, 56, 24)   96
block_2_project[0][0]
__________________________________________________________________________________________________
block_2_add (Add)               (None, 56, 56, 24)   0
block_1_project_BN[0][0]
block_2_project_BN[0][0]
__________________________________________________________________________________________________
block_3_expand (Conv2D)         (None, 56, 56, 144)  3456
block_2_add[0][0]
__________________________________________________________________________________________________
block_3_expand_BN (BatchNormali (None, 56, 56, 144)  576
block_3_expand[0][0]
__________________________________________________________________________________________________
block_3_expand_relu (ReLU)      (None, 56, 56, 144)  0
block_3_expand_BN[0][0]
__________________________________________________________________________________________________
block_3_pad (ZeroPadding2D)     (None, 57, 57, 144)  0
block_3_expand_relu[0][0]
__________________________________________________________________________________________________
block_3_depthwise (DepthwiseCon (None, 28, 28, 144)  1296
block_3_pad[0][0]
__________________________________________________________________________________________________
```

```
block_3_depthwise_BN (BatchNorm (None, 28, 28, 144)  576
block_3_depthwise[0][0]
__________________________________________________________________________________________________
block_3_depthwise_relu (ReLU)   (None, 28, 28, 144)  0
block_3_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_3_project (Conv2D)        (None, 28, 28, 32)   4608
block_3_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_3_project_BN (BatchNormal (None, 28, 28, 32)   128
block_3_project[0][0]
__________________________________________________________________________________________________
block_4_expand (Conv2D)         (None, 28, 28, 192)  6144
block_3_project_BN[0][0]
__________________________________________________________________________________________________
block_4_expand_BN (BatchNormali (None, 28, 28, 192)  768
block_4_expand[0][0]
__________________________________________________________________________________________________
block_4_expand_relu (ReLU)      (None, 28, 28, 192)  0
block_4_expand_BN[0][0]
__________________________________________________________________________________________________
block_4_depthwise (DepthwiseCon (None, 28, 28, 192)  1728
block_4_expand_relu[0][0]
__________________________________________________________________________________________________
block_4_depthwise_BN (BatchNorm (None, 28, 28, 192)  768
block_4_depthwise[0][0]
__________________________________________________________________________________________________
block_4_depthwise_relu (ReLU)   (None, 28, 28, 192)  0
block_4_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_4_project (Conv2D)        (None, 28, 28, 32)   6144
block_4_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_4_project_BN (BatchNormal (None, 28, 28, 32)   128
block_4_project[0][0]
__________________________________________________________________________________________________
```

```
block_4_add (Add)               (None, 28, 28, 32)   0
block_3_project_BN[0][0]
block_4_project_BN[0][0]
__________________________________________________________________________________________________
block_5_expand (Conv2D)         (None, 28, 28, 192)  6144
block_4_add[0][0]
__________________________________________________________________________________________________
block_5_expand_BN (BatchNormali (None, 28, 28, 192)  768
block_5_expand[0][0]
__________________________________________________________________________________________________
block_5_expand_relu (ReLU)      (None, 28, 28, 192)  0
block_5_expand_BN[0][0]
__________________________________________________________________________________________________
block_5_depthwise (DepthwiseCon (None, 28, 28, 192)  1728
block_5_expand_relu[0][0]
__________________________________________________________________________________________________
block_5_depthwise_BN (BatchNorm (None, 28, 28, 192)  768
block_5_depthwise[0][0]
__________________________________________________________________________________________________
block_5_depthwise_relu (ReLU)   (None, 28, 28, 192)  0
block_5_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_5_project (Conv2D)        (None, 28, 28, 32)   6144
block_5_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_5_project_BN (BatchNormal (None, 28, 28, 32)   128
block_5_project[0][0]
__________________________________________________________________________________________________
block_5_add (Add)               (None, 28, 28, 32)   0
block_4_add[0][0]
block_5_project_BN[0][0]
__________________________________________________________________________________________________
block_6_expand (Conv2D)         (None, 28, 28, 192)  6144
block_5_add[0][0]
__________________________________________________________________________________________________
block_6_expand_BN (BatchNormali (None, 28, 28, 192)  768
block_6_expand[0][0]
```

```
----------------------------------------------------------------------------------
------------------
block_6_expand_relu (ReLU)      (None, 28, 28, 192)  0
block_6_expand_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_6_pad (ZeroPadding2D)     (None, 29, 29, 192)  0
block_6_expand_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_6_depthwise (DepthwiseCon (None, 14, 14, 192)  1728
block_6_pad[0][0]
----------------------------------------------------------------------------------
------------------
block_6_depthwise_BN (BatchNorm (None, 14, 14, 192)  768
block_6_depthwise[0][0]
----------------------------------------------------------------------------------
------------------
block_6_depthwise_relu (ReLU)   (None, 14, 14, 192)  0
block_6_depthwise_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_6_project (Conv2D)        (None, 14, 14, 64)   12288
block_6_depthwise_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_6_project_BN (BatchNormal (None, 14, 14, 64)   256
block_6_project[0][0]
----------------------------------------------------------------------------------
------------------
block_7_expand (Conv2D)         (None, 14, 14, 384)  24576
block_6_project_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_7_expand_BN (BatchNormali (None, 14, 14, 384)  1536
block_7_expand[0][0]
----------------------------------------------------------------------------------
------------------
block_7_expand_relu (ReLU)      (None, 14, 14, 384)  0
block_7_expand_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_7_depthwise (DepthwiseCon (None, 14, 14, 384)  3456
block_7_expand_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_7_depthwise_BN (BatchNorm (None, 14, 14, 384)  1536
block_7_depthwise[0][0]
```

```
----------------------------------------------------------------------------------
------------------
block_7_depthwise_relu (ReLU)   (None, 14, 14, 384)  0
block_7_depthwise_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_7_project (Conv2D)        (None, 14, 14, 64)   24576
block_7_depthwise_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_7_project_BN (BatchNormal (None, 14, 14, 64)   256
block_7_project[0][0]
----------------------------------------------------------------------------------
------------------
block_7_add (Add)               (None, 14, 14, 64)   0
block_6_project_BN[0][0]
block_7_project_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_8_expand (Conv2D)         (None, 14, 14, 384)  24576
block_7_add[0][0]
----------------------------------------------------------------------------------
------------------
block_8_expand_BN (BatchNormali (None, 14, 14, 384)  1536
block_8_expand[0][0]
----------------------------------------------------------------------------------
------------------
block_8_expand_relu (ReLU)      (None, 14, 14, 384)  0
block_8_expand_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_8_depthwise (DepthwiseCon (None, 14, 14, 384)  3456
block_8_expand_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_8_depthwise_BN (BatchNorm (None, 14, 14, 384)  1536
block_8_depthwise[0][0]
----------------------------------------------------------------------------------
------------------
block_8_depthwise_relu (ReLU)   (None, 14, 14, 384)  0
block_8_depthwise_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_8_project (Conv2D)        (None, 14, 14, 64)   24576
block_8_depthwise_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_8_project_BN (BatchNormal (None, 14, 14, 64)   256
```

```
block_8_project[0][0]
__________________________________________________________________________________________________
block_8_add (Add)               (None, 14, 14, 64)   0
block_7_add[0][0]
block_8_project_BN[0][0]
__________________________________________________________________________________________________
block_9_expand (Conv2D)         (None, 14, 14, 384)  24576
block_8_add[0][0]
__________________________________________________________________________________________________
block_9_expand_BN (BatchNormali (None, 14, 14, 384)  1536
block_9_expand[0][0]
__________________________________________________________________________________________________
block_9_expand_relu (ReLU)      (None, 14, 14, 384)  0
block_9_expand_BN[0][0]
__________________________________________________________________________________________________
block_9_depthwise (DepthwiseCon (None, 14, 14, 384)  3456
block_9_expand_relu[0][0]
__________________________________________________________________________________________________
block_9_depthwise_BN (BatchNorm (None, 14, 14, 384)  1536
block_9_depthwise[0][0]
__________________________________________________________________________________________________
block_9_depthwise_relu (ReLU)   (None, 14, 14, 384)  0
block_9_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_9_project (Conv2D)        (None, 14, 14, 64)   24576
block_9_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_9_project_BN (BatchNormal (None, 14, 14, 64)   256
block_9_project[0][0]
__________________________________________________________________________________________________
block_9_add (Add)               (None, 14, 14, 64)   0
block_8_add[0][0]
block_9_project_BN[0][0]
__________________________________________________________________________________________________
block_10_expand (Conv2D)        (None, 14, 14, 384)  24576
block_9_add[0][0]
__________________________________________________________________________________________________
```

```
------------------
block_10_expand_BN (BatchNormal (None, 14, 14, 384)  1536
block_10_expand[0][0]
--------------------------------------------------------------------------------
------------------
block_10_expand_relu (ReLU)     (None, 14, 14, 384)  0
block_10_expand_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_10_depthwise (DepthwiseCo (None, 14, 14, 384)  3456
block_10_expand_relu[0][0]
--------------------------------------------------------------------------------
------------------
block_10_depthwise_BN (BatchNor (None, 14, 14, 384)  1536
block_10_depthwise[0][0]
--------------------------------------------------------------------------------
------------------
block_10_depthwise_relu (ReLU)  (None, 14, 14, 384)  0
block_10_depthwise_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_10_project (Conv2D)       (None, 14, 14, 96)   36864
block_10_depthwise_relu[0][0]
--------------------------------------------------------------------------------
------------------
block_10_project_BN (BatchNorma (None, 14, 14, 96)   384
block_10_project[0][0]
--------------------------------------------------------------------------------
------------------
block_11_expand (Conv2D)        (None, 14, 14, 576)  55296
block_10_project_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_11_expand_BN (BatchNormal (None, 14, 14, 576)  2304
block_11_expand[0][0]
--------------------------------------------------------------------------------
------------------
block_11_expand_relu (ReLU)     (None, 14, 14, 576)  0
block_11_expand_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_11_depthwise (DepthwiseCo (None, 14, 14, 576)  5184
block_11_expand_relu[0][0]
--------------------------------------------------------------------------------
------------------
block_11_depthwise_BN (BatchNor (None, 14, 14, 576)  2304
block_11_depthwise[0][0]
--------------------------------------------------------------------------------
```

```
------------------
block_11_depthwise_relu (ReLU)  (None, 14, 14, 576)  0
block_11_depthwise_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_11_project (Conv2D)       (None, 14, 14, 96)   55296
block_11_depthwise_relu[0][0]
--------------------------------------------------------------------------------
------------------
block_11_project_BN (BatchNorma (None, 14, 14, 96)   384
block_11_project[0][0]
--------------------------------------------------------------------------------
------------------
block_11_add (Add)              (None, 14, 14, 96)   0
block_10_project_BN[0][0]
block_11_project_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_12_expand (Conv2D)        (None, 14, 14, 576)  55296
block_11_add[0][0]
--------------------------------------------------------------------------------
------------------
block_12_expand_BN (BatchNormal (None, 14, 14, 576)  2304
block_12_expand[0][0]
--------------------------------------------------------------------------------
------------------
block_12_expand_relu (ReLU)     (None, 14, 14, 576)  0
block_12_expand_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_12_depthwise (DepthwiseCo (None, 14, 14, 576)  5184
block_12_expand_relu[0][0]
--------------------------------------------------------------------------------
------------------
block_12_depthwise_BN (BatchNor (None, 14, 14, 576)  2304
block_12_depthwise[0][0]
--------------------------------------------------------------------------------
------------------
block_12_depthwise_relu (ReLU)  (None, 14, 14, 576)  0
block_12_depthwise_BN[0][0]
--------------------------------------------------------------------------------
------------------
block_12_project (Conv2D)       (None, 14, 14, 96)   55296
block_12_depthwise_relu[0][0]
--------------------------------------------------------------------------------
------------------
block_12_project_BN (BatchNorma (None, 14, 14, 96)   384
block_12_project[0][0]
```

```
----------------------------------------------------------------------------------
------------------
block_12_add (Add)              (None, 14, 14, 96)   0
block_11_add[0][0]
block_12_project_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_13_expand (Conv2D)        (None, 14, 14, 576)  55296
block_12_add[0][0]
----------------------------------------------------------------------------------
------------------
block_13_expand_BN (BatchNormal (None, 14, 14, 576)  2304
block_13_expand[0][0]
----------------------------------------------------------------------------------
------------------
block_13_expand_relu (ReLU)     (None, 14, 14, 576)  0
block_13_expand_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_13_pad (ZeroPadding2D)    (None, 15, 15, 576)  0
block_13_expand_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_13_depthwise (DepthwiseCo (None, 7, 7, 576)    5184
block_13_pad[0][0]
----------------------------------------------------------------------------------
------------------
block_13_depthwise_BN (BatchNor (None, 7, 7, 576)    2304
block_13_depthwise[0][0]
----------------------------------------------------------------------------------
------------------
block_13_depthwise_relu (ReLU)  (None, 7, 7, 576)    0
block_13_depthwise_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_13_project (Conv2D)       (None, 7, 7, 160)    92160
block_13_depthwise_relu[0][0]
----------------------------------------------------------------------------------
------------------
block_13_project_BN (BatchNorma (None, 7, 7, 160)    640
block_13_project[0][0]
----------------------------------------------------------------------------------
------------------
block_14_expand (Conv2D)        (None, 7, 7, 960)    153600
block_13_project_BN[0][0]
----------------------------------------------------------------------------------
------------------
block_14_expand_BN (BatchNormal (None, 7, 7, 960)    3840
```

```
block_14_expand[0][0]
__________________________________________________________________________________________________
block_14_expand_relu (ReLU)     (None, 7, 7, 960)    0
block_14_expand_BN[0][0]
__________________________________________________________________________________________________
block_14_depthwise (DepthwiseCo (None, 7, 7, 960)    8640
block_14_expand_relu[0][0]
__________________________________________________________________________________________________
block_14_depthwise_BN (BatchNor (None, 7, 7, 960)    3840
block_14_depthwise[0][0]
__________________________________________________________________________________________________
block_14_depthwise_relu (ReLU)  (None, 7, 7, 960)    0
block_14_depthwise_BN[0][0]
__________________________________________________________________________________________________
block_14_project (Conv2D)       (None, 7, 7, 160)    153600
block_14_depthwise_relu[0][0]
__________________________________________________________________________________________________
block_14_project_BN (BatchNorma (None, 7, 7, 160)    640
block_14_project[0][0]
__________________________________________________________________________________________________
block_14_add (Add)              (None, 7, 7, 160)    0
block_13_project_BN[0][0]
block_14_project_BN[0][0]
__________________________________________________________________________________________________
block_15_expand (Conv2D)        (None, 7, 7, 960)    153600
block_14_add[0][0]
__________________________________________________________________________________________________
block_15_expand_BN (BatchNormal (None, 7, 7, 960)    3840
block_15_expand[0][0]
__________________________________________________________________________________________________
block_15_expand_relu (ReLU)     (None, 7, 7, 960)    0
block_15_expand_BN[0][0]
__________________________________________________________________________________________________
block_15_depthwise (DepthwiseCo (None, 7, 7, 960)    8640
block_15_expand_relu[0][0]
__________________________________________________________________________________________________
```

```
block_15_depthwise_BN (BatchNor (None, 7, 7, 960)    3840
block_15_depthwise[0][0]
__________________________________________________________________________________
__________________
block_15_depthwise_relu (ReLU)  (None, 7, 7, 960)    0
block_15_depthwise_BN[0][0]
__________________________________________________________________________________
__________________
block_15_project (Conv2D)       (None, 7, 7, 160)    153600
block_15_depthwise_relu[0][0]
__________________________________________________________________________________
__________________
block_15_project_BN (BatchNorma (None, 7, 7, 160)    640
block_15_project[0][0]
__________________________________________________________________________________
__________________
block_15_add (Add)              (None, 7, 7, 160)    0
block_14_add[0][0]
block_15_project_BN[0][0]
__________________________________________________________________________________
__________________
block_16_expand (Conv2D)        (None, 7, 7, 960)    153600
block_15_add[0][0]
__________________________________________________________________________________
__________________
block_16_expand_BN (BatchNormal (None, 7, 7, 960)    3840
block_16_expand[0][0]
__________________________________________________________________________________
__________________
block_16_expand_relu (ReLU)     (None, 7, 7, 960)    0
block_16_expand_BN[0][0]
__________________________________________________________________________________
__________________
block_16_depthwise (DepthwiseCo (None, 7, 7, 960)    8640
block_16_expand_relu[0][0]
__________________________________________________________________________________
__________________
block_16_depthwise_BN (BatchNor (None, 7, 7, 960)    3840
block_16_depthwise[0][0]
__________________________________________________________________________________
__________________
block_16_depthwise_relu (ReLU)  (None, 7, 7, 960)    0
block_16_depthwise_BN[0][0]
__________________________________________________________________________________
__________________
block_16_project (Conv2D)       (None, 7, 7, 320)    307200
block_16_depthwise_relu[0][0]
__________________________________________________________________________________
```

```
------------------
block_16_project_BN (BatchNorma (None, 7, 7, 320)    1280
block_16_project[0][0]
--------------------------------------------------------------------------------
------------------
Conv_1 (Conv2D)                 (None, 7, 7, 1280)   409600
block_16_project_BN[0][0]
--------------------------------------------------------------------------------
------------------
Conv_1_bn (BatchNormalization)  (None, 7, 7, 1280)   5120        Conv_1[0][0]
--------------------------------------------------------------------------------
------------------
out_relu (ReLU)                 (None, 7, 7, 1280)   0           Conv_1_bn[0][0]
================================================================================
==================
Total params: 2,257,984
Trainable params: 2,223,872
Non-trainable params: 34,112
--------------------------------------------------------------------------------
------------------
```

[42]:
```
tf.config.experimental_run_functions_eagerly(True)
```

[11]:
```
from skimage.transform import resize
from skimage.color import gray2rgb
import numpy as np
from tensorflow import keras


class ImageGenerator(keras.utils.Sequence):
    # after https://stanford.edu/~shervine/blog/
 ↪keras-how-to-generate-data-on-the-fly
    def __init__(self, train_x, train_y, shuffle=True, batch_size=10):
        self.train_x = train_x
        self.train_y = train_y
        self.shuffle = shuffle
        self.batch_size = batch_size
        self.on_epoch_end()

    def __len__(self):
        return len(self.train_x) // self.batch_size

    def __getitem__(self, index):
        batch_ids = self.indexes[index*self.batch_size:(index+1)*self.
 ↪batch_size]
        X, y = self.__data_generation(batch_ids)
        return X, y
```

```
    def on_epoch_end(self):
        self.indexes = np.arange(len(self.train_x))
        if self.shuffle == True:
            np.random.shuffle(self.indexes)

    def __data_generation(self, batch_ids):
        X = []
        y = []
        for id in batch_ids:
            X.append(imagenet_transform(self.train_x[id, ...]))
            y.append(self.train_y[id, ...])

        return np.array(X), np.array(y)

def imagenet_transform(train_image):
    image_resized = resize(
        train_image, (224, 224),
        anti_aliasing=True
    )
    return gray2rgb(image_resized)
```

[37]:
```
train_images[0, ...].shape
```

[37]: (28, 28)

[12]:
```
imagenet_transform(train_images[0:1])
```

[12]:
```
array([[[[0., 0., 0.],
         [0., 0., 0.],
         [0., 0., 0.],
         …,
         [0., 0., 0.],
         [0., 0., 0.],
         [0., 0., 0.]],

        [[0., 0., 0.],
         [0., 0., 0.],
         [0., 0., 0.],
         …,
         [0., 0., 0.],
         [0., 0., 0.],
         [0., 0., 0.]],

        [[0., 0., 0.],
         [0., 0., 0.],
         [0., 0., 0.],
```

```
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 …,

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]]],


[[[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],
```

```
 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 …,

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]]],


[[[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
```

```
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 …,

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]]],


…,


[[[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
```

```
 [0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.]],

[[0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.],
 …,
 [0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.]],

[[0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.],
 …,
 [0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.]],

…,

[[0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.],
 …,
 [0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.]],

[[0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.],
 …,
 [0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.]],

[[0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.],
 …,
 [0., 0., 0.],
 [0., 0., 0.],
 [0., 0., 0.]]],
```

```
[[[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 …,

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
```

```
  [0., 0., 0.],
  [0., 0., 0.]]],


[[[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 …,

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.],
  …,
  [0., 0., 0.],
  [0., 0., 0.],
  [0., 0., 0.]],

 [[0., 0., 0.],
```

```
        [0., 0., 0.],
        [0., 0., 0.],
        …,
        [0., 0., 0.],
        [0., 0., 0.],
        [0., 0., 0.]]]])
```

[35]:
```
def create_transfer_model():
    base_model = tf.keras.applications.MobileNetV2(
        input_shape=(224, 224, 3),
        include_top=False,
        weights='imagenet'
    )
    base_model.trainable = False
    model = tf.keras.Sequential([
      base_model,
      tf.keras.layers.GlobalAveragePooling2D(),
      tf.keras.layers.Dense(len(class_names))
    ])
    compile_model(model)
    return model
```

[36]:
```
model = create_transfer_model()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applicatio
ns/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h
5
9406464/9406464             0s
0us/step
```

Model: "sequential_4"

```
 Layer (type)                        Output Shape                          ␣
↪Param #

 mobilenetv2_1.00_224 (Functional)   (None, 7, 7, 1280)                   ␣
↪2,257,984

 global_average_pooling2d            (None, 1280)                              ␣
↪  0
 (GlobalAveragePooling2D)                                                      ␣
↪

 dense_9 (Dense)                     (None, 10)                            ␣
↪12,810
```

Total params: 2,270,794 (8.66 MB)

Trainable params: 12,810 (50.04 KB)

Non-trainable params: 2,257,984 (8.61 MB)

[41]:
```
# model.fit(  # if you use tensorflow < 2.20, you should use fit_generator␣
 ↪instead
#     x=ImageGenerator(train_images, train_labels),
#     epochs=50,
#     verbose=1,
#     validation_data=ImageGenerator(test_images, test_labels)
# )

from tensorflow.keras import layers, models

model = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(32, 3, activation='relu'),
    layers.MaxPooling2D(),
    layers.Conv2D(64, 3, activation='relu'),
    layers.MaxPooling2D(),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

model.fit(
    train_images[..., None],
    train_labels,
    batch_size=10,
    epochs=10,
    validation_data=(test_images[..., None], test_labels)
)
```

```
Epoch 1/10
6000/6000              34s 5ms/step -
accuracy: 0.8488 - loss: 0.4123 - val_accuracy: 0.8826 - val_loss: 0.3265
```

```
Epoch 2/10
6000/6000              31s 5ms/step -
accuracy: 0.8960 - loss: 0.2801 - val_accuracy: 0.8957 - val_loss: 0.2847
Epoch 3/10
6000/6000              32s 5ms/step -
accuracy: 0.9128 - loss: 0.2369 - val_accuracy: 0.9069 - val_loss: 0.2644
Epoch 4/10
6000/6000              31s 5ms/step -
accuracy: 0.9216 - loss: 0.2063 - val_accuracy: 0.9044 - val_loss: 0.2661
Epoch 5/10
6000/6000              32s 5ms/step -
accuracy: 0.9318 - loss: 0.1813 - val_accuracy: 0.9086 - val_loss: 0.2627
Epoch 6/10
6000/6000              32s 5ms/step -
accuracy: 0.9389 - loss: 0.1600 - val_accuracy: 0.9102 - val_loss: 0.2774
Epoch 7/10
6000/6000              32s 5ms/step -
accuracy: 0.9461 - loss: 0.1409 - val_accuracy: 0.9047 - val_loss: 0.3086
Epoch 8/10
6000/6000              35s 6ms/step -
accuracy: 0.9516 - loss: 0.1257 - val_accuracy: 0.9105 - val_loss: 0.3096
Epoch 9/10
6000/6000              35s 6ms/step -
accuracy: 0.9568 - loss: 0.1119 - val_accuracy: 0.9099 - val_loss: 0.3265
Epoch 10/10
6000/6000              37s 6ms/step -
accuracy: 0.9619 - loss: 0.0999 - val_accuracy: 0.9107 - val_loss: 0.3430
```

[41]:
```
<keras.src.callbacks.history.History at 0x151b329eb10>
```

[ ]:
```
```